

Why D_4

Specificity of the carrier against matched comparison lattices (R95)

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Why D_4

Abstract

FFGFT posits the lattice D_4 as the carrier of the compact geometry $T^4/\mathbb{Z}_3$ (Doc. 314) **[SETZUNG]**. Doc. 190 lists under R95 as an open bridge that an adversarial specificity test against matched comparison carriers is missing. This document examines the carrier. Three criteria fixed in advance are applied to $\mathbb{Z}^4$, A_4 , $A_2 \oplus A_2$, D_4 and to random lattices: the fixed-point-free $\mathbb{Z}_3$ action with nine fixed points carried in the corpus (Doc. 330), the kissing number, and the lattice energy as the Epstein zeta function at equal volume of the fundamental cell. Only the third criterion is independent of FFGFT. The $\mathbb{Z}_3$ condition excludes $\mathbb{Z}^4$ and A_4 ; among the remaining carriers D_4 has the lowest lattice energy **[K]**, and the kissing number 24 is consistent with Doc. 340. D_4 is thereby better motivated as a carrier but remains a posit; R95 is not closed, and the level of the operator $\hat{F}$ remains open **[S]**. Verification script: 13/13.

Status markers

[B]	Algebraically proved.
[K]	Numerically confirmed (verification script).
[S]	Structurally plausible, open.
[Q]	Taken from the literature, cited.
[SETZUNG]	Axiomatic posit.

1 The question

A posit is motivated rather than arbitrary if the chosen carrier stands out among comparable alternatives by quantities that were not tuned to the results. This is exactly what the GAE logic demands, as applied in the HLV audit: there an ordinary cubic lattice outperformed the carrier under study in the spectral measure, and the specificity claim failed (Doc. 330). For D_4 , R95 records that $5 \nmid |\text{Aut}(D_4)| = 1152$ is a lattice-specific fact, but not a proof of specificity.

The test must therefore meet three conditions. The comparison carriers have the same dimension and the same volume of the fundamental cell. The criteria are fixed before the calculation. And a criterion meant to establish specificity must not be a number that originates in FFGFT. Criteria that express a corpus condition test only compatibility; they are explicitly marked as such below.

2 Comparison carriers and criteria

The comparison covers the cubic lattice $\mathbb{Z}^4$ (the winner in the HLV audit), the root lattice A_4 , the sum $A_2 \oplus A_2$ and D_4 , each scaled to unit volume of the fundamental cell, plus random lattices.

K1: $\mathbb{Z}_3$ compatibility (corpus condition). The corpus carries $T^4/\mathbb{Z}_3$ with nine fixed points and a $\mathbb{Z}_3$ action with eigenvalues $\{\omega, \omega^2, \omega, \omega^2\}$; the nine fixed points follow from $|\det(1 - g)| = |(1 - \omega)(1 - \omega^2)|^2 = 9$ (Doc. 330, likewise Docs. 361, 364, 365). What is required is therefore a lattice automorphism g of order 3 without eigenvalue 1. K1 expresses this condition and tests compatibility, not specificity. The search is complete, because every automorphism maps shortest vectors to shortest vectors and all four bases consist of shortest vectors.

Which $\mathbb{Z}_3$ is meant is shown by the complete enumeration of the 1152 automorphisms of D_4 : 80 have order 3. Of these, 64 have a two-dimensional fixed space and produce fixed tori instead of isolated fixed points, among them the Dynkin automorphism of triality. The remaining 16 are fixed-point-free with exactly nine fixed points on T^4 [K]. These are the 16 elements named in Doc. 330; the triality that carries the orbifold is thus this fixed-point-free class, not the Dynkin automorphism.

K2: Kissing number (consistency). The number of shortest lattice vectors. It is a property of the lattice alone, but FFGFT uses it itself ($\text{Kissing}(D_4) = 24$ in Doc. 340). K2 therefore tests consistency, not specificity.

K3: Lattice energy. The Epstein zeta function

$$Z_\Lambda(v) = \sum_{v \in \Lambda \setminus \{0\}} |v|^{-v} \quad (1)$$

at unit volume of the fundamental cell, for $\nu = 5, 6, 8$. It is at the same time the spectral zeta function of the flat torus of the dual lattice, hence an operator quantity of the Laplacian, and it contains no free parameter. It is computed with EpsteinLib (Buchheit et al.) [Q]; the library reproduces the closed-form values for $\mathbb{Z}^4$ and D_4 to twelve digits [K].

3 Results

K1. $\mathbb{Z}^4$ and A_4 have no automorphism of order 3 without eigenvalue 1; on them $T^4/\mathbb{Z}_3$ with nine fixed points is impossible [K]. $A_2 \oplus A_2$ and D_4 have one, each with $|\det(1 - g)| = 9$ [K]. The winner of the HLV audit is thus already excluded by the symmetry condition.

K2. The kissing numbers are D_4 : 24, A_4 : 20, $A_2 \oplus A_2$: 12, $\mathbb{Z}^4$: 8 [K]. Of the two $\mathbb{Z}_3$ -compatible carriers, only D_4 has the 24 that enters Doc. 340 as $\text{Kissing}(D_4)$. This is consistency: the corpus computes with the lattice it posits.

K3.

ν	D_4	A_4	$A_2 \oplus A_2$	$\mathbb{Z}^4$
5	22.860	23.123	23.703	24.531
6	12.583	12.919	13.699	14.830
8	6.830	7.282	8.457	10.246

D_4 has the lowest energy for all three exponents [K]. Of 300 general random lattices none falls below D_4 (best 13.15 at $\nu = 6$), and likewise none of 300 random lattices from the $\mathbb{Z}_3$ -compatible family (best 12.72) [K]. All 200 small perturbations around D_4 increase the energy; D_4 is a local minimum [K].

It is proved that D_4 is a local minimum of the Epstein zeta function among all four-dimensional lattices (Sarnak and Strömbergsson 2006) [Q]. Whether it is also the global minimum is an open conjecture in four dimensions. The 600 random lattices support it but do not prove it.

4 Assessment

What is shown. Among the four-dimensional carriers that admit the $\mathbb{Z}_3$ action with nine fixed points carried in the corpus, D_4 has the lowest lattice energy [K]. This is the only one of the three quantities that does not originate in FFGFT; the kissing number confirms consistency only. D_4 is thereby better motivated as a carrier than before, but it remains a motivated posit [SETZUNG]. The finding does not replace a proof of specificity.

What remains open. The GAE logic also asks about the operator: whether $\hat{F}$ with $\xi = \lambda_{\min}(\hat{F}_{D_4})$ (Doc. 327) yields something distinguishable on the comparison carriers. This is not tested here [S]. Likewise, the random search is numerical evidence, not a proof of global minimality within the $\mathbb{Z}_3$ family; it covers the exponents $\nu = 5, 6, 8$.

Consequences for the bookkeeping. R95 is not closed. The required operator test is still outstanding, and new results receive no register entry.

5 Summary

Criterion	D_4	$A_2 \oplus A_2$	A_4	$\mathbb{Z}^4$
K1: $\mathbb{Z}_3$ with 9 fixed points (corpus)	yes	yes	no	no
K2: kissing number (consistency)	24	12	20	8
K3: $Z(6)$ at unit volume	12.58	13.70	12.92	14.83

Among the carriers that satisfy the symmetry condition, D_4 has the lowest lattice energy, a quantity that does not originate in FFGFT; the kissing number is consistent with the corpus. The posit of the carrier is thus better supported, but it remains a posit.

Verification script. 2/python/Dok380_Skripte/pruef_380_d4_spezifitaet.py (13/13): check of the library against closed-form values, K1 to K3, classification of the order-3 automorphisms of D_4 , random and perturbation tests.

Sources. Buchheit, A. A.; Busse, J. K.; Gutendorf, R.: *Computation and Properties of the Epstein Zeta Function with Applications to Quantum Systems*. arXiv:2412.16317 (2024); software EpsteinLib, <https://github.com/epsteinlib/epsteinlib>. Sarnak, P.; Strömbergsson, A.: *Minima of Epstein's zeta function and heights of flat tori*. Invent. Math. **165**, 115–151 (2006). Conway, J. H.; Sloane, N. J. A.: *Sphere Packings, Lattices and Groups*. 3rd ed., Springer (1999).